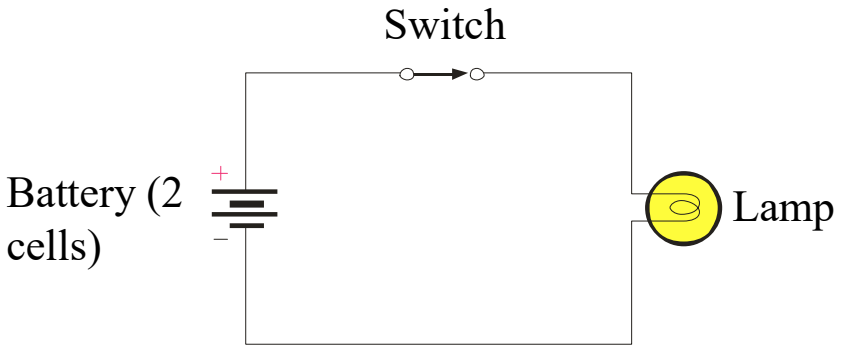
Outline

- Introduction to Electronics
- Voltage, Current, Resistance
- Ohm's Law

What is Electronics?

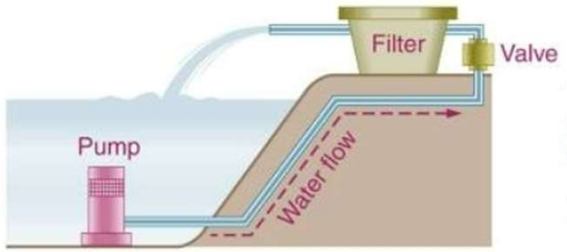
- Electronics is the study and use of devices that control the flow of electrical current.
- Deals with signals, circuits, and components.



A simple electric circuit

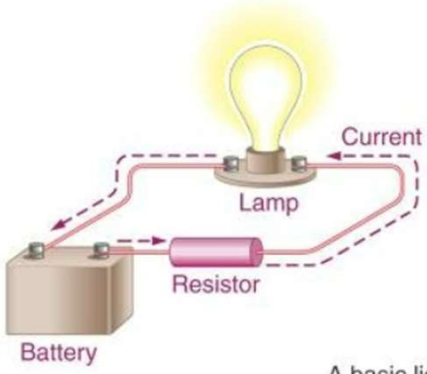
Basic Electrical Quantities

The Oldest Analogy in Electronics



The **pump** provides the **hydraulic force** that pushes **water** through the **pipe**. The rate of **water flow** depends on the pressure provided by the pump, and cannot exceed the capacity of the pipe. The **valve** is used to set the water flow to some desired rate.

A water garden filtration system



The **battery** provides the **electromotive force (EMF)** that pushes **charge** through the **wire**. The **current** depends on the EMF provided by the battery, and cannot exceed the capacity of the wire. The **resistor** is used to set the current to some desired rate.

A basic lighting circuit

- **Voltage (V):** Electrical “pressure” (push).
- **Current (I):** Flow of electrons (like water flow).
- **Resistance (R):** Opposition to flow.
- **Power (P):** Energy used (P = V × I).

Voltage, Current and Resistance

- **Voltage (V):**
  - A **"difference of potential"** or the source that generates a **flow of charge (current)** through a circuit.
  - Unit of Voltage is volt (V)

- **Current (I):**
  - Electric Current is the controlled flow of electrons through a conductor
  - The unit of Current is Ampere (A):

- **Resistance (R)**
  - **Resistance is the** opposition to current
  - Unit of Resistance – Ohm (Ω - Greek letter omega)

Ohm's Law

The most important fundamental law in electronics is **Ohm's law**, which relates voltage, current, and resistance.

I = the circuit current
V = the applied voltage
R = the circuit resistance

Formula: **V = I × R**

- *Current is **directly** proportional to **voltage***
- *Current is **inversely** proportional to **resistance***

$$V = IR$$

$$I = \frac{V}{R}$$

Basic Electrical Quantities - Units

Electrical Property	Units	Description
Current (I)	Ampere (A)	The rate of flow of electrical charge
Voltage (E or V)	Volts (V)	Electric Force
Resistance (R)	Ohm (Ω)	The opposition to current.